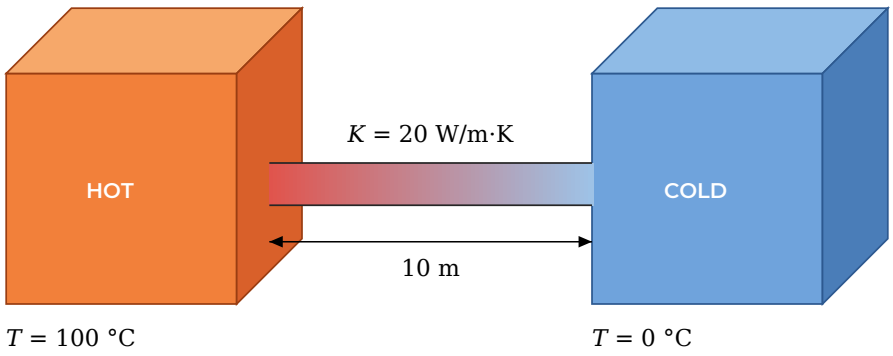
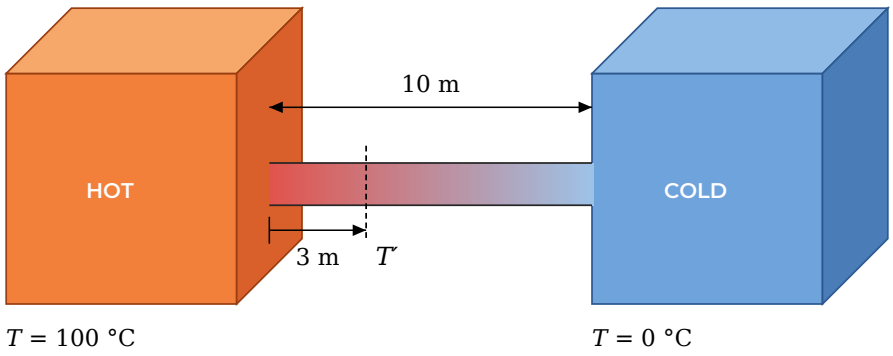
Q1. If a $10\ \text{m}$ long rod of cross-sectional area $A = 1\ \text{cm}^2$ has thermal conductivity $K = 20\ \text{W/m·K}$ as shown, find out the amount of energy transferred through the rod in $5\ \text{s}$.



Solution

Q2. If a $10\ \mathrm{m}$ long rod of cross-sectional area $A = 1\ \mathrm{cm}^2$ has thermal conductivity $K = 20\ \mathrm{W/m{\cdot}K}$ as shown, find the temperature T' of the cross-section of the rod at $3\ \mathrm{m}$ away from the hotter end.

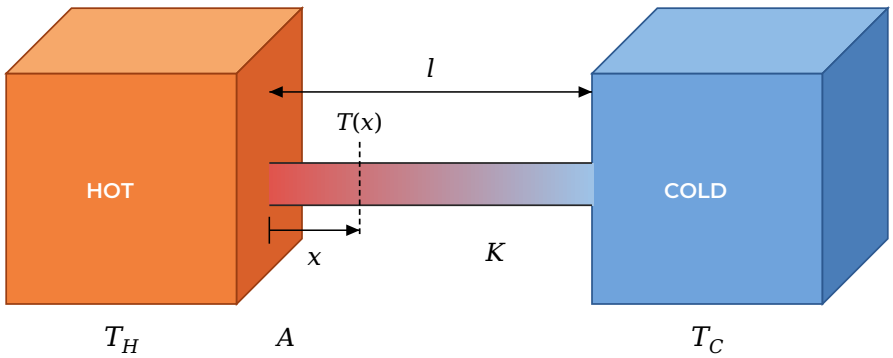


Solution

Q3. For the given system,

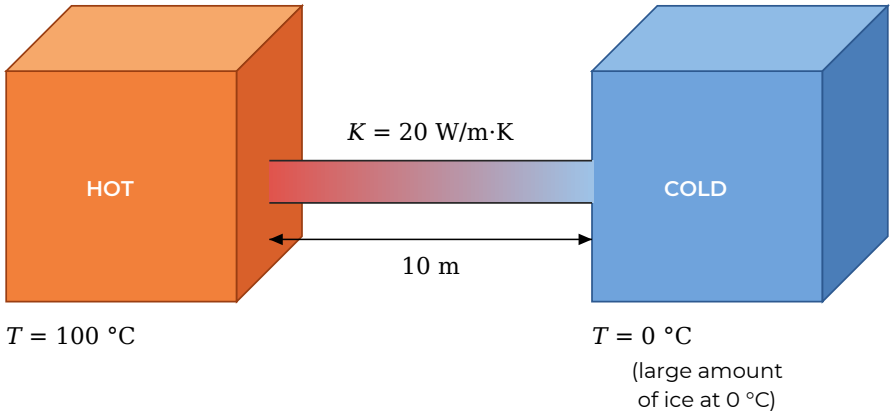
(a) Find out the temperature as a function of x.

(b) Draw the graph of temperature as a function of x.



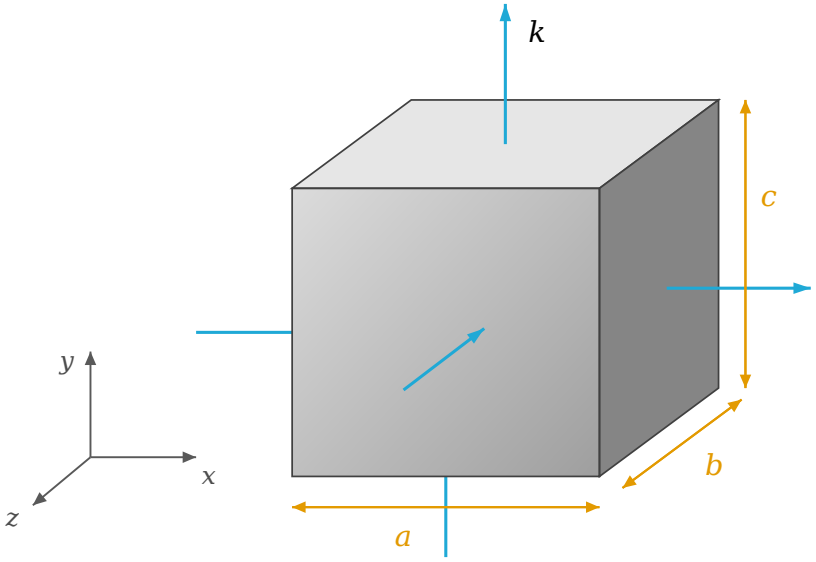
Solution

Q4. If a $10\ \text{m}$ long rod of cross-sectional area $A = 1\ \text{cm}^2$ has thermal conductivity $K = 20\ \text{W/m·K}$ as shown, find out the amount of ice that will melt in $1\ \text{hour}$.



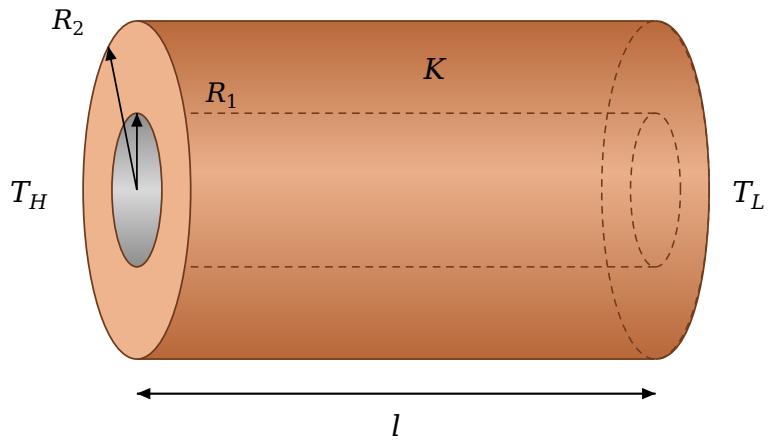
Solution

Q5. If a cuboid of dimensions (a, b, c) shown has thermal conductivity k, find out the thermal resistance of the cuboid along perpendicular to its surfaces.



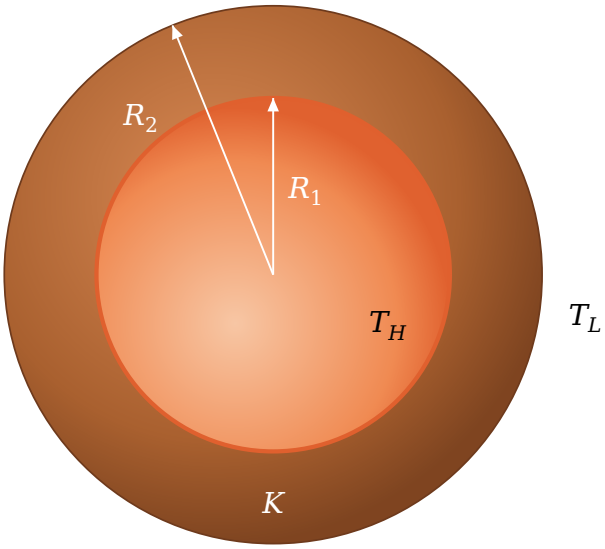
Solution

Q6. A cylinder of inner radius R_1, outer radius R_2 and length l has thermal conductivity K as shown. If its one end is at temperature T_H and another end at temperature T_L, then find out its thermal resistance (R_T).



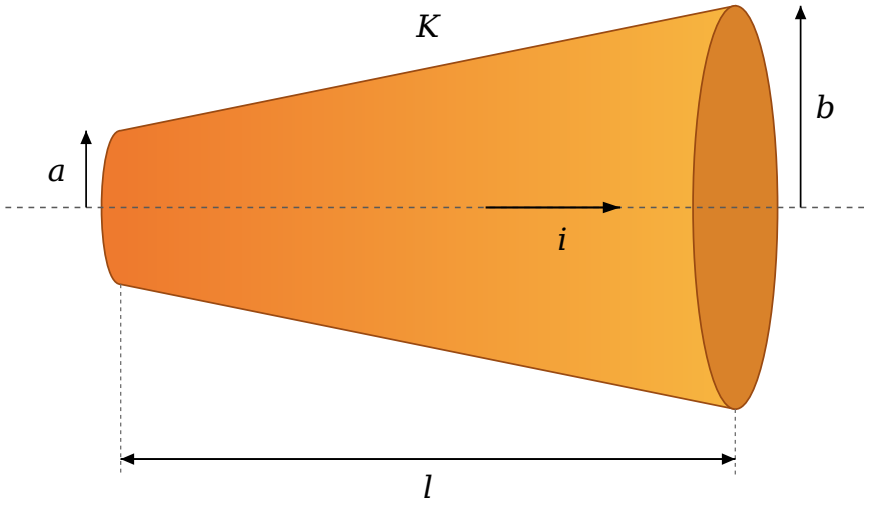
Solution

Q7. A sphere of inner radius R_1 and outer radius R_2 has thermal conductivity K as shown. If its inner surface is at temperature T_H and outer surface at temperature T_L, then find out its thermal resistance (R_T).



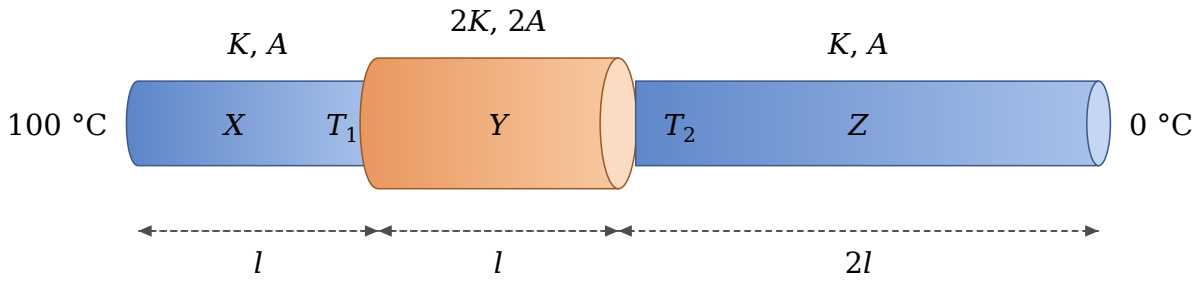
Solution

Q8. A frustum of length l has radii a and b such that $(b - a) \ll l$. If its thermal conductivity is K and the heat current through it is i as shown, then find out its thermal resistance (R_T).



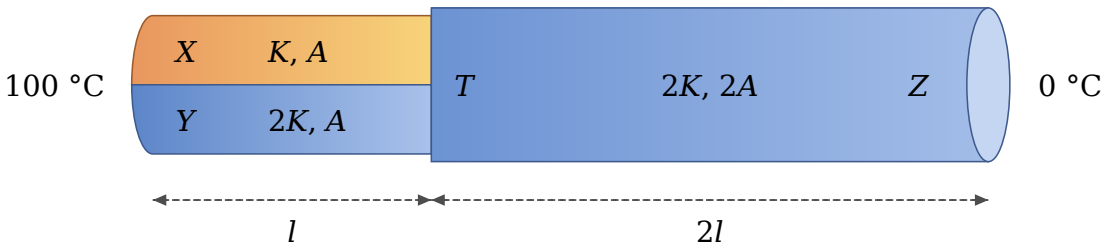
Solution

Q9. Three rods X, Y and Z are connected in series as shown. One of the ends of the rods X and Z are exposed to the atmosphere at temperature $100\ °C$ and $0\ °C$, respectively. Find out the temperatures T_1 at the interface of X and Y and T_2 at the interface of Y and Z. Assume no heat loss to the surrounding.



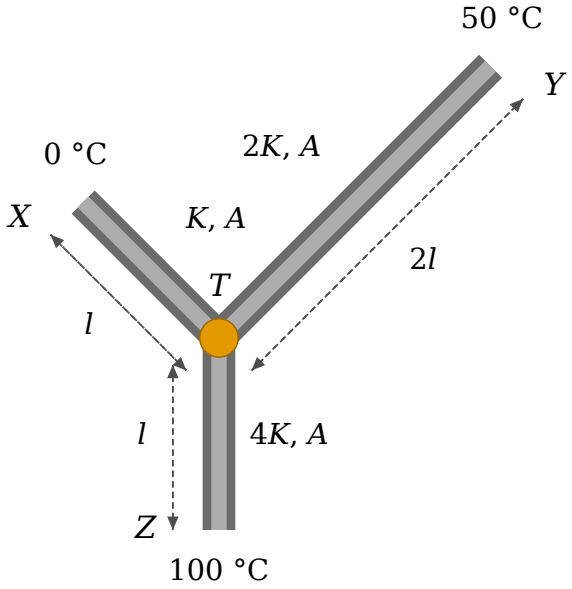
Solution

Q10. Three rods X, Y and Z are connected as shown. One of the ends of the rods X, Y and Z are exposed to the atmosphere at temperature 100 °C and 0 °C. Find out the temperature T at the interface of X, Y and Z. Assume no heat loss to the surrounding.



Solution

Q11. Three rods X, Y and Z are connected as shown. Find out the temperature T at the junction of the rods if other end of the rods are exposed to the atmosphere at temperature $0\ °C$, $50\ °C$ and $100\ °C$, respectively. Assume no heat loss to the surrounding.

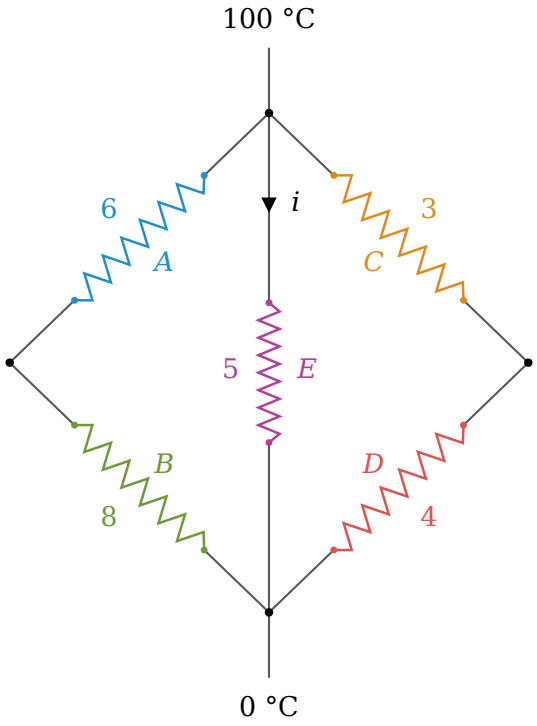


Solution

Q12. Five thermal resistors A, B, C, D and E are connected as shown. The temperature at the junction of thermal resistors A, C and E is 100 °C and B, D and E is 0 °C.

(a) Is this a Wheatstone bridge?

(b) Find out the thermal current i through the thermal resistor E.



Solution

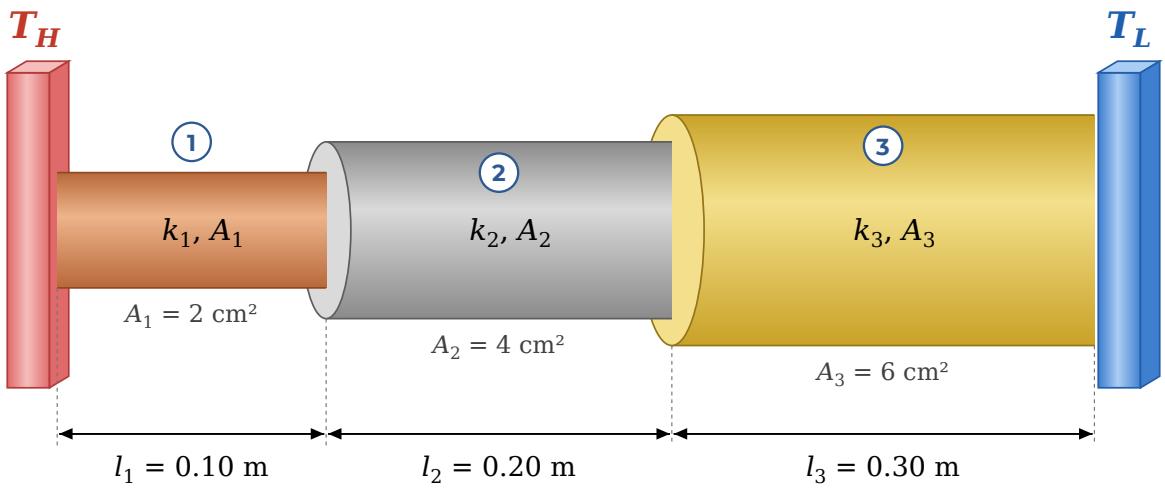
Q13. Three conducting sections are connected **in series** between a hot and a cold reservoir.

Section 1: $l_1 = 0.10\ \text{m},\ A_1 = 2\ \text{cm}^2,\ k_1 = 100\ \text{W m}^{-1}\ \text{K}^{-1}$

Section 2: $l_2 = 0.20\ \text{m},\ A_2 = 4\ \text{cm}^2,\ k_2 = 50\ \text{W m}^{-1}\ \text{K}^{-1}$

Section 3: $l_3 = 0.30\ \text{m},\ A_3 = 6\ \text{cm}^2,\ k_3 = ?$

The effective thermal conductivity of the complete combination is $K_E = 60\ \text{W m}^{-1}\ \text{K}^{-1}$. Find the thermal conductivity k_3 of the third section.



Solution

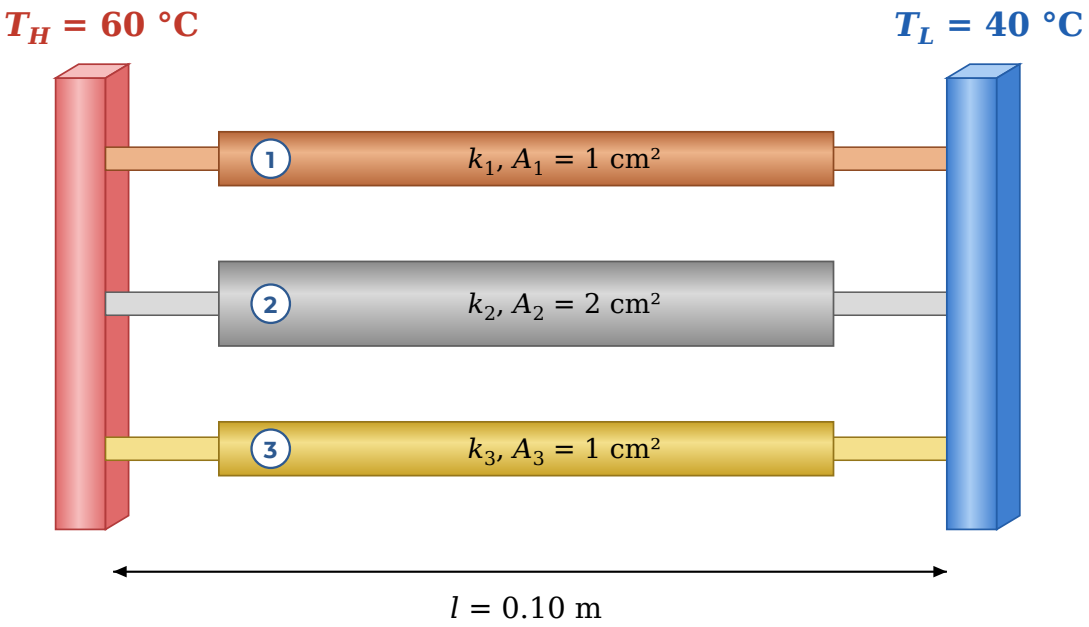
Q14. Three rods are connected **in parallel** between two plates maintained at $T_H = 60\ °\text{C}$ and $T_L = 40\ °\text{C}$. All three rods have the same length $l = 0.10\ \text{m}$. Their properties are:

Rod 1: $k_1 = 200\ \text{W m}^{-1}\ \text{K}^{-1},\ A_1 = 1\ \text{cm}^2$

Rod 2: $k_2 = 100\ \text{W m}^{-1}\ \text{K}^{-1},\ A_2 = 2\ \text{cm}^2$

Rod 3: $k_3 = 100\ \text{W m}^{-1}\ \text{K}^{-1},\ A_3 = 1\ \text{cm}^2$

Find the effective thermal conductivity K_E of the parallel combination.



Solution